

# 2025 Ch H2 Q1

**Section:** Chemical Changes and Structure

**Topic:** Structure and Bonding

## Question Summary:

This question is about ionisation energies, ionic size and ionic bonding. You are asked to: (1) write an equation for the first ionisation energy of sodium, (2) use ionisation energy data to calculate the energy needed to form  $\text{Ca}^{2+}(\text{g})$ , (3) explain differences in ionisation energies for sodium and magnesium, (4) explain the relative sizes of  $\text{Na}^+$  and  $\text{Mg}^{2+}$  ions, and (5) apply the idea of ionic bonding and moles to a potassium intake calculation.

## Worked Solution

(a)(i) First ionisation of sodium

The first ionisation energy is the energy needed to remove one mole of electrons from one mole of gaseous atoms to form one mole of gaseous ions.

For sodium, the equation is:



(a)(ii) Energy required for  $\text{Ca}(\text{g}) \rightarrow \text{Ca}^{2+}(\text{g}) + 2\text{e}^-$

The process  $\text{Ca}(\text{g}) \rightarrow \text{Ca}^{2+}(\text{g}) + 2\text{e}^-$  involves removing two electrons, so you add the first and second ionisation energies of calcium.

From the table:

First ionisation energy of Ca =  $590 \text{ kJ mol}^{-1}$

Second ionisation energy of Ca =  $1145 \text{ kJ mol}^{-1}$

Total energy =  $590 + 1145 = 1735 \text{ kJ mol}^{-1}$

(a)(iii) Second ionisation energy of sodium vs magnesium

After losing one electron, sodium forms  $\text{Na}^+$  with the same electron arrangement as neon. The next electron to be removed (for the second ionisation) would come from a full inner shell which is much closer to the nucleus and more strongly attracted. This makes the second ionisation energy of sodium very high.

Magnesium forms  $\text{Mg}^+$  after losing one electron, but  $\text{Mg}^+$  still has one electron left in the outer 3s subshell. The second ionisation of magnesium removes this remaining outer-shell electron, which is less strongly attracted to the nucleus than an inner-shell electron, so the second ionisation energy of magnesium is much lower than that of sodium.

(a)(iv) Why  $\text{Mg}^{2+}$  is smaller than  $\text{Na}^+$

Both  $\text{Na}^+$  and  $\text{Mg}^{2+}$  ions have the same number of occupied energy levels (they each have the electron arrangement of neon). However,  $\text{Mg}^{2+}$  has 12 protons in the nucleus, while  $\text{Na}^+$  has 11 protons.

The higher nuclear charge in  $\text{Mg}^{2+}$  pulls the same number of electrons closer to the nucleus. With similar shielding but a larger nuclear charge, the attraction between the nucleus and the electrons is stronger in  $\text{Mg}^{2+}$ , so its ionic radius is smaller than that of  $\text{Na}^+$ .

(b)(i) Meaning of ionic bond

An ionic bond is the electrostatic attraction between positively charged ions and negatively charged ions.

(b)(ii) Potassium tablets and bananas

Step 1: Calculate the total moles of potassium taken in tablet form.

Eight tablets per day for seven days gives:

Number of tablets =  $8 \times 7 = 56$

Each tablet contains 0.0012 moles of potassium.

Total moles of potassium =  $56 \times 0.0012 = 0.0672 \text{ mol}$

Step 2: Convert moles of potassium to mass using GFM =  $39.1 \text{ g mol}^{-1}$ .

Mass of potassium =  $0.0672 \text{ mol} \times 39.1 \text{ g mol}^{-1} = 2.62752 \text{ g}$

Step 3: Calculate the number of bananas that give this mass of potassium.

Each banana contains  $450 \text{ mg} = 0.450 \text{ g}$  of potassium.

Number of bananas =  $2.62752 \text{ g} / 0.450 \text{ g} = 5.84\dots$

You must round up to the next whole banana to provide at least the same mass of potassium.

So the adult would need 6 bananas.

## Final Answers

(a)(i)  $\text{Na(g)} \rightarrow \text{Na}^+(\text{g}) + \text{e}^-$

(a)(ii)  $1735 \text{ kJ mol}^{-1}$

(a)(iii) Sodium's second ionisation removes an inner-shell electron from  $\text{Na}^+$ , so it is much higher than magnesium's second ionisation, which removes an outer-shell electron from  $\text{Mg}^+$ .

(a)(iv)  $\text{Mg}^{2+}$  is smaller than  $\text{Na}^+$  because it has a greater nuclear charge but the same number of occupied energy levels, so electrons are held more tightly.

(b)(i) Ionic bond: electrostatic attraction between positive and negative ions.

(b)(ii) Number of whole bananas required = 6.

## Revision Tips

- Learn the definition of first ionisation energy and practise writing correct equations with state symbols.
- Remember that big jumps in successive ionisation energies show when an inner-shell electron is being removed.
- Compare ions by thinking about both nuclear charge and number of occupied energy levels.
- Definition of ionic bond must include the idea of electrostatic attraction and oppositely charged ions.
- In mole calculations, keep track of steps: moles  $\rightarrow$  mass, and always round up when asked for whole objects like bananas.